Analysis and Design of Algorithms

NUS National University of Singapore

Algorithms

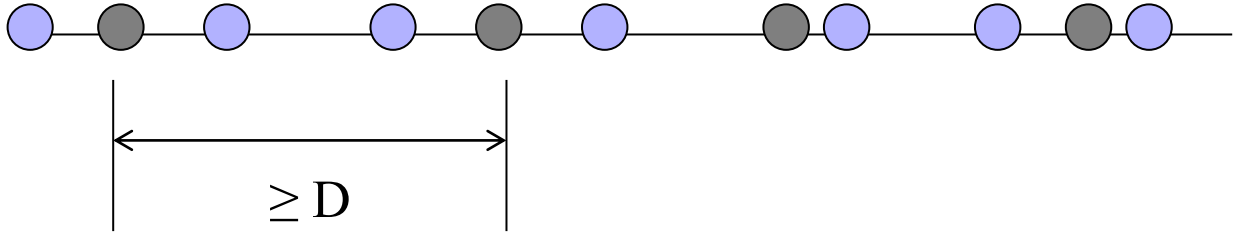
CS3230

Tutorial

Week 11

Question 4

YuckDonald wants to open as many of its chain restaurant on Orchard Road as possible. It has found n suitable locations for its restaurants $a_1, \ldots, a_n$. The restaurants should be at least D distance apart so that they do not compete with each other.



Question 4(a)

Describe how the problem can be modeled as a maximum independent set problem, where an independent set in a graph $G = (V, E)$ is a subset of vertices V such that no two vertices in the graph is connected by an edge.

Question 4(b)

Look for optimal substructure in YuckDonalds problem and argue that

$$M(V) = 1 + M(V - \{a_j \in V : d(a_i, a_j) < D\})$$

where $M(V)$ is the size of the largest set of restaurants that can fit in V, a_i is an element in an optimal solution and $d(a_i, a_j)$ is the distance between a_i and a_j.

Question 4(c)

Assume that the locations for a set of restaurant $a_1, \ldots, a_n$ is sorted and lie on a straight line. Show that a_1 is part of some optimal solution, i.e. selecting the smallest element satisfies the greedy choice property.

Question 4(d)

Complete your greedy algorithm for evaluating $M(V)$ and argue its correctness by mathematical induction.